

Pablo Ibáñez-Porras

Predoctoral Researcher at CISA-INIA-CSIC

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PROFESSIONAL PROFILE

Predoctoral researcher specialising in spatial epidemiology, mathematical modelling, data science and geographic information systems for One Health surveillance and risk assessment.

RESEARCH PROJECT EXPERIENCE

- 2026 - 2027** **Development of a model for the spread of HPAI among wild birds within the Iberian Peninsula and identification of epidemiological links between cases through phylogenetic analysis.**
Animal Health Research Centre (CISA-INIA-CSIC)
Position funded through a management agreement with the Spanish Ministry of Agriculture, Fisheries and Food (2025 - 2027)
- 2024 - 2026** **Development of models, viewers and applications for rapid risk assessment of emerging pathogens**
Animal Health Research Centre (CISA-INIA-CSIC)
Position funded by the European Partnership on Animal Health and Welfare (2024 - 2027)
- 2023 - 2024** **Cost-benefit model of early wild-boar vaccination following African swine fever (ASF) outbreaks**
Animal Health Research Centre (CISA-INIA-CSIC)
Position funded by the VACDIVA project (2023 - 2024)
- 2022 - 2023** **Enhancement of the DiFLusion warning system for avian-influenza introduction by wild birds in Spain and development of ProtectIA to assess introduction risk in poultry holdings**
Animal Health Research Centre (CISA-INIA-CSIC)
Position funded through a management agreement with the Spanish Ministry of Agriculture, Fisheries and Food (2022 - 2023)

EDUCATION

- 2025 - present** **PhD in Computer Engineering (ongoing)**
University of Salamanca
- 2021** **Master's Degree in Mathematical Modelling**
University of Salamanca
- 2021** **Monte Carlo modelling of proton-beam interactions with different targets**
Centre for Pulsed, Ultrashort and Ultraintense Lasers (CLPU)
- 2020** **BSc in Physics**
University of Salamanca

SCIENTIFIC PUBLICATIONS

- Abdrakhmanov, S.K., Iglesias, I., Ibáñez-Porras, P., Mukhanbetkaliyev, Y.Y., Mukhanbetkaliyeva, A.A., Ruzmatov, S.I., Kadyrov, A.S., Balgabekova, K.T. (2026). Development of a dashboard for monitoring and risk assessment of highly pathogenic avian influenza in Kazakhstan. *Central Asian Journal of Veterinary Science*. [https://doi.org/10.51452/cajvs.2026.2\(014\).2224](https://doi.org/10.51452/cajvs.2026.2(014).2224)
- Duarte-Benvenuto A, Sacristán C, Ewbank AC, Costa-Silva S, Groch KR, Díaz-Delgado J, Ibáñez-Porras P, Colosio AC, da Cunha Gomes Ramos H, Carvalho VL, Ribeiro VL, Bertozzi CP, Pessi CF, Gravena W, da Silva VMF, Marmontel M, Marigo J, Catão-Dias JL. (2026). Molecular surveillance of flaviviruses in wild cetaceans from Brazil. *Journal of Wildlife Diseases*. <https://doi.org/10.7589/JWD-D-25-00149>
- Ibáñez-Porras, P., Bosch, J., Feliziani, F., Maresca, C., Sanchez-Vizcaino, J. M., Iglesias, I., et al., Martinez Aviles, M. (2025). WiBISS: A tool to estimate avoided lost revenue of African swine fever wild boar vaccination at municipality level. *Frontiers in Veterinary Science*. <https://doi.org/10.3389/fvets.2025.1667173>
- Iglesias, I., Ruzmatov, S., Ibáñez-Porras, P., Kadyrov, A., Bakishev, T., Korennoy, F., Perez, A., Abdrakhmanov, S. (2025). Risk Assessment of Highly Pathogenic Avian Influenza (HPAI) in Wild Bird Species in Kazakhstan. *German Journal of Veterinary Research*. <https://doi.org/10.51585/gjvr.2025.3.0149>
- Ibáñez-Porras, P., de la Torre, A., Gómez-Pérez, J. I., Tomás-Tenllado, C., García, E., Cáceres, G., Pérez, A., Iglesias, I. (2024). DiFLusion: A novel space-time alert system for HPAI. *German Journal of Veterinary Research*. <https://doi.org/10.51585/gjvr.2024.4.0107>
- Sacristán C, Ewbank AC, Ibáñez Porras P, Pérez Ramírez E, de la Torre A, Briones V, Iglesias I. (2024). Novel epidemiologic features of highly pathogenic avian influenza virus A H5N1 2.3.3.4b panzootic: a review. *Transboundary and Emerging Diseases*. <https://doi.org/10.1155/2024/5322378>

Ewbank AC, Catão-Dias JL, Navas-Suarez PE, Duarte-Benvenuto A, Zamana-Ramblas R, Ferreira-Machado E, Christino Lial H, Ibañez-Porras P, Sacristán I, Sacristán C. (2024). Novel alpha-, beta-, and gammaherpesviruses in Neotropical carnivores of Brazil.. *Transboundary and Emerging Diseases*. <https://doi.org/10.1155/2024/1347516>

Duarte-Benvenuto A, Sánchez-Sarmiento AM, Ewbank AC, Zamana-Ramblas R, Costa-Silva S, Silvestre N, Fanta T, Keid LB, Soares RM, Pessi CF, Sabbadini JR, Borges MF, Ferioli RB, Marcon M, Barbosa CB, Fernandes NCCA, Ibáñez-Porras P, Navas-Suárez PE, Catão-Dias JL, Sacristán C. (2024). Bacterial septicemia and herpesvirus infection in Antarctic fur seals (*Arctocephalus gazella*) stranded in the São Paulo coast, Brazil.. *Veterinary Research Communications*. <https://doi.org/10.1007/s11259-024-10408-x>

Duarte-Benvenuto A, Sacristán C, Ewbank AC, Zamana-Ramblas R, Lial HC, Silva SC, Arias Lugo MA, Keid LB, Pessi CF, Sabbadini JR, Ribeiro VL, do Valle RDR, Bertozzi CP, Colosio AC, Ramos HDCG, Sánchez-Sarmiento AM, Ferioli RB, Pavanelli L, Ikeda JMP, Carvalho VL, Catardo Gonçalves FA, Ibáñez-Porras P, Sacristán I, Catão-Dias JL. (2023). Molecular Detection and Characterization of *Mycoplasma* spp. in Marine Mammals from Brazil.. *Emerging Infectious Diseases*. <https://doi.org/10.3201/eid2912.23090>

TECHNICAL AND KNOWLEDGE-TRANSFER ARTICLES

Iglesias Irene; Lim Seunghyun,. de la Torre Ana, Kanankege Kaushi, Ibañez Pablo, Fillman Kathryn, Blanco Carlos E, Perez Andres M (2023). DashFLUboard: a web-based tool for assessing and predicting global spread of highly pathogenic avian influenza in near real time. *EsriNews*. [Link](#)

Irene Iglesias, Elena García, Germán Cáceres, Pablo Ibañez, José Ignacio Gómez, Christian Tomas-Tenllado, Carlos Eduardo Blanco Urbina, Ana de la Torre (2023). Influenza Aviar: cambios en la dinámica de la enfermedad y cómo anticiparnos a ella mediante sistemas de vigilancia a tiempo real. *Avinews*. [Link](#)

SCIENTIFIC ACTIVITY

26 scientific events recorded · 20 lead-author or speaker contributions · 14 international contributions.

Lead-author contributions

- SMART-E: A Spatial Digital Framework for Standardized Environmental AMR Surveillance - Oral; European One Health Association Annual Scientific Meeting 2026 (EOHA 2026) (2026)
- From Science to Policy: Co-developing a Real-Time One Health Early Warning System for Avian Influenza (DiFLUtion) - Poster; European One Health Association Annual Scientific Meeting 2026 (EOHA 2026) (2026)
- ASF FrontWave: A Rapid Risk Assessment interactive mapping tool - Poster; SVEPM Conference 2026+ (2026)
- Mapping anthropogenic sources of antimicrobial-resistance emissions - Oral; Esri Spain Conference 2026 (2025)
- ProtectIA: A geospatial tool for avian-influenza prevention - Oral; 4th International Young Researchers Meeting INNOVA-Salamanca (2025)
- Geoprocessing tools to support emerging-disease response - Oral; 1st Interdisciplinary Congress on Medicine and Engineering (2025)
- WiBISS: A tool to estimate economic benefits of African swine fever wild boar vaccination for pig producers - Poster; GARA Scientific Meeting (2025)
- DiFLUtion: a tool for surveillance against Avian Influenza - Oral; 2nd General Assembly of the CSIC Computational Biology and Bioinformatics Network (Conexión BCB) (2025)
- Epidemiology and spatial analysis: using GIS to understand disease dynamics and manage public and animal health from a One Health perspective - Oral; Esri Spain Conference 2024 (2024)
- Soil contamination by antibiotics and antimicrobial-resistance genes - Poster; 10th CONDEGRES Symposium (2024)
- MarineStrandingViewer: An Online Tool to Record Marine Mammal Strandings - Poster; IAAAM 55th Annual Conference (2024)
- Real-time avian-influenza warning system: DiFLUtion - Poster; Health Surveillance Meeting 2023: present and future towards 2030 (2023)
- Ants, chickens and mathematics - Oral; INIAciando ciencia (2023)
- DiFLUtion avian-influenza surveillance system: direct use of data from SEO/BirdLife citizen-science programmes - Poster; Spanish Ornithology Congress (2022)

TEACHING, ADVICE AND KNOWLEDGE TRANSFER

- The overlooked role of Passeriformes in avian influenza and antimicrobial resistance amid climate change - Workshop LINCGLOBAL Scientific and Communication Training (2025)
- Training course on Risk analysis for transboundary animal disease control - World Organization for Animal Health (WOAH) Sub-Regional Representation for South-East Asia (2025)
- Implementing AI in animal-health research - Workshop on Artificial Intelligence, Big Data and digitalisation in veterinary medicines (2025)
- Mentor for the Scientists in Practice programme - AEAC / CSIC - CISA (2024)
- Analysis of the spatio-temporal distribution of several animal diseases - External-placement supervision (2023)
- Mathematical tools for temporal smoothing of migration routes - Bachelor's thesis supervision (2023)
- Analysis of the spatio-temporal distribution of HPAI outbreaks in Spain - Universidad Complutense de Madrid (2022)
- Technical advice on the highly pathogenic avian influenza (HPAI) situation and the use of DiFLUtion as an early-warning system (2022-2023).

SELECTED CONTINUING EDUCATION

Selected courses from 607 accredited hours of continuing education.

- Biodiversity analysis with QGIS, Google Earth Engine and OpenData - 125 h · GeoInnova
- Advanced Python - 40 h · CSIC
- Geoprocessing scripts with Python in ArcGIS Pro - 40 h · Esri España
- Machine learning for scientific research: random forests, boosting and advanced interpretation - 24 h · Museo Nacional de Ciencias Naturales (MNCN-CSIC)
- Deep learning with Keras for scientific research - 21 h · Museo Nacional de Ciencias Naturales (MNCN-CSIC)

- Web development with the ArcGIS API for JavaScript - 25 h · Esri España
- Services with Docker - 20 h · CSIC (Plan de Formación)
- Adapted numerical methods for differential equations - 10 h · University of Salamanca
- FME and databases - 8 h · Consejo Superior de Investigaciones Científicas (CSIC)

DEVELOPED TOOLS

- DiFLUtion - Spatio-temporal warning system for highly pathogenic avian influenza
- ProtectIA - Multi-criteria geospatial tool for avian-influenza prevention
- SMART-E - Surveillance and Monitoring of Antimicrobial Resistance in the Environment
- WiBISS - Wild Boar Immunization Strategy Simulator
- ASF FrontWave - African Swine Fever Front-Wave Rapid Risk Assessment
- Dashboards - Epidemiological dashboards for surveillance and risk assessment

SCIENCE OUTREACH

- Epidemic detectives: chickens and mathematics - Festival Pint of Science (2025)
- Mentor for the Scientists in Practice programme - (2024)
- Epidemics and pandemics: become the most dangerous virus - Semana de la Ciencia y la Tecnología (2022)
- Become a real epidemic detective - Semana de la Ciencia y la Tecnología (2022)
- Epidemics and pandemics: become the most dangerous virus - Noche Europea de los Investigadores (2022)

AWARDS, FELLOWSHIPS AND RESEARCH STAYS

- MOVTEC 2025-12 research stay · Consejo Superior de Investigaciones Científicas (CSIC) · 2026
- 2024 Research Award · Instituto de Estudios del Huevo · 2024
- Introduction to Research Fellowship · Instituto Universitario de Física Fundamental y Matemáticas (IUFFYM) · 2021

SKILLS

- Python, MATLAB, Mathematica and Fortran
- QGIS, ArcGIS Pro/Online, Google Earth Engine and remote sensing
- Mathematical modelling, spatio-temporal analysis and machine learning
- Docker, geospatial application development and data visualisation
- LaTeX and Microsoft Office
- Professional English